THE PALAEOMAGNETISM OF THE SUDBURY DIABASE DYKE SWARM

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Previously published results on the palaeomagnetism of the Sudbury diabase dyke swarm had indicated that these dykes were formed at low palaeomagnetic latitude. A new and independent set of data described in the present study confirms this view and suggests further that the dykes were not disturbed significantly since the time of their intrusion, over 1 000 m.y. ago. It is also suggested that the dykes sampled were intruded intermittently over a period extending for less than 10 000 years.

Les données paléomagnétiques sur les dykes de Sudbury publiées antérieurement à cette étude s'accordaient pour indiquer que ceux-ci ont vraisemblablement été injectés près de l'équateur paléomagnétique. La nouvelle série de résultats présentée dans ce rapport confirme cette hypothèse et suggère en plus que les dykes n'ont pas été dérangés de façon appréciable depuis l'époque de leur injection, soit depuis au delà d'un milliard d'années. On croit pouvoir déduire aussi de ces résultats que les dykes échantillonnés ont été formés par intermittence au cours d'une période de beaucoup inférieure à 10 000 ans.

INTRODUCTION

Before the present study, three independent sets of results (Strangway 1961, 1964; Sopher 1963; and Fahrig *et al.* 1965) were published on the palaeomagnetism of the northwesterly trending diabase dykes that profusely intrude the rocks of the Sudbury irruptive and of its surroundings. These dykes are commonly known to be olivine-bearing, differing in that respect from other important diabase dyke swarms elsewhere in the Canadian shield. Detailed petrological descriptions of the dykes have been given by Kindle (1933) and by Collins (1935), among others. The width of a dyke may be anywhere between a few inches and 400 ft and some of the dykes may extend for scores of miles. The dykes are undisputedly the youngest rock unit in the area and several radiogenic age determinations (Fairbairn, Hurley, and Pinson 1960; Fahrig and Wanless 1963; Van Schmus 1964) indicate that they may be from 1 000 to 1 800 m.y. old. It has been suggested by Kindle (1933) that the dykes were intruded in tensional fractures, although it is not known whether their intrusion took place as a simultaneous event or intermittently over a long period of time. Another unknown is the amount of tectonic deformation that they may have undergone since the time of their original cooling.

It is interesting to note (Table I) that the mean directions of magnetization for each of the sets of palaeomagnetic data mentioned above are in strikingly good agreement among themselves. This gives a fairly good idea of the general attitude of the earth's field in the area at the time the dykes were intruded, but it gives no indication of their geological history. Moreover, since two of the three results are based on relatively highly scattered directions of magnetization, the precision of the overall result is necessarily low and the exactness of the conclusions that might be derived from it would be impaired by this fact.

TABLE I

Summary of previous palaeomagnetic directions

S	*N*	*k*	*D*	*I*	Demag. field	Reference
1	8	5.1	265	8	100–200	Strangway (1961, 1964)
4	7	31.1	260	−2	250	Sopher (1963)
22	37	4.1	264	1	130	Fahrig *et al.* (1965)
Mean			263	+2		

NOTE: *k*, precision estimate (Fisher, 1953). *S*, number of sites sampled. *N*, number of samples. *D*, clockwise angle between astronomic north and palaeomagnetic meridian. *I*, inclination of magnetic vector below horizontal plane.

The present study was undertaken as an attempt to evaluate the relative importance of various possible sources of magnetization scattering in the swarm and perhaps to shed some light on its geological history.

SAMPLING AND MEASUREMENTS

A total of 62 independently oriented cores was drilled from 16 sites distributed along 13 different dykes southwest of the Sudbury Basin (Fig. 1). The

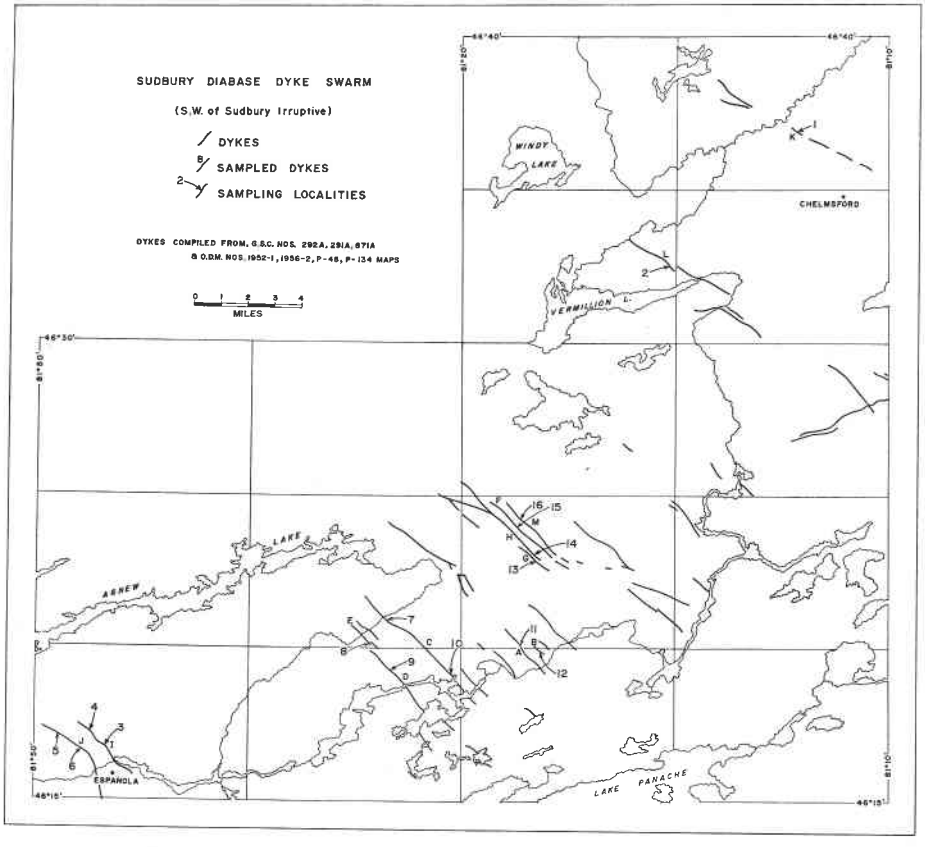


FIG. 1. Map showing sampling localities and sampled dykes.

sampling and measuring procedures and instrumentations were essentially the same as described in a previous paper (Larochelle 1966). The angle θ between the directions of magnetization of the two cylinders cut from each core was used as a criterion for magnetization homogeneity; cores for which this angle exceeded 23° were not considered homogeneously magnetized. The tolerance limit of 23° was set for a minimum within-core precision k of 25, as explained earlier (Larochelle 1966). In testing the magnetic stability of the dykes, spot samples were submitted to successive alternating field cleaning treatments in demagnetizing fields of 50, 100, 150 . . . up to 700 oersteds (peak) by steps of 50 oersteds. The instrumentation and technique for this treatment were described in detail in a previous paper (Larochelle and Black 1965). In the light of these tests the rest of the cores were treated in a field of 150 oersteds (peak).

Magnetically unstable cores were characterized by a sharp decrease in magnetization intensity accompanied by an increase of θ beyond 23° after the treatment in 150 oersteds. Subsequent treatments in higher fields resulted in increases of the value of θ generally beyond 30° and in sluggish decreases of the intensity of magnetization (Fig. 2). In all, eight cores were found to be

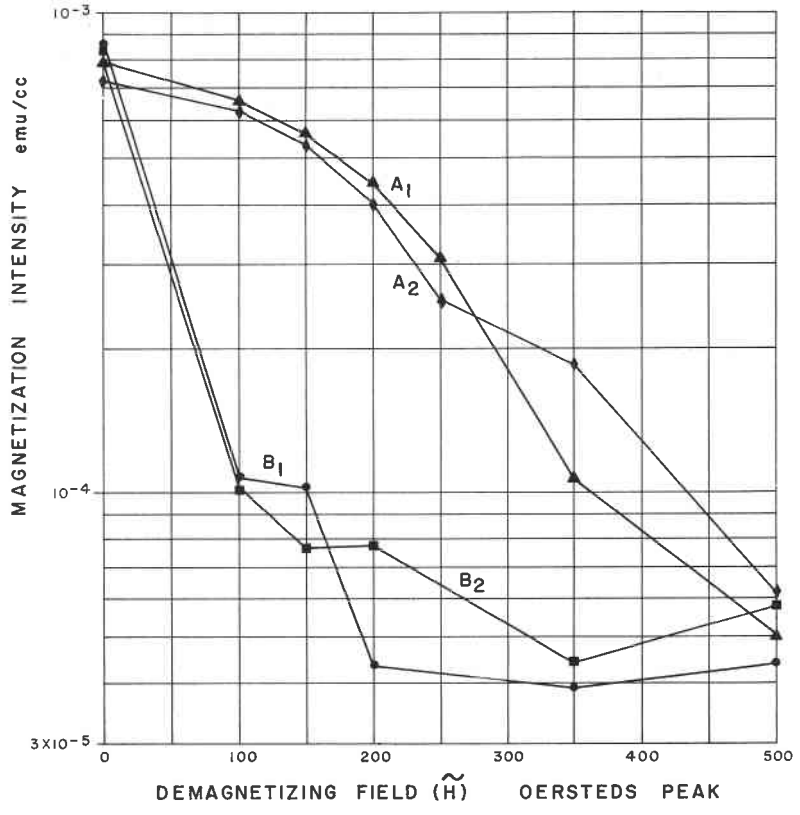


FIG. 2. Demagnetization curves of typical stable (A) and unstable (B) cores. Subscripts refer to cylinders Nos. 1 and 2 for each core respectively.

magnetically unstable. Their magnetic instability seemed to be related to their texture and (or) to the composition of their ferromagnetic ingredients, as indicated by the fact that six out of the eight unstable cores had been collected at two sites where the mineralogical composition and texture would be expected to be fairly consistent.

Only one core displayed any signs of having been remagnetized by lightning. Its natural remanent magnetization was unusually resistant to cleaning treatments in low fields and, upon further treatments in higher fields, it lost its homogeneity, although it remained at least 20 times stronger than that of any of the other cores treated in such fields.

The magnetization directions of the nine cores just discussed were not considered as reflecting the attitude of the earth's field at the time the dykes were formed, and accordingly will not be taken into account in the rest of this analysis. On the other hand, the magnetization of one core, which was inhomogeneous before the cleaning treatment, became homogeneous after the first and further treatments. Its magnetic stability and homogeneity justified including it as a significant sample.

Mean directions of magnetization at each site, before and after the 150 oersted treatment, are listed in Table II, along with the corresponding Fisher precision estimate k of the statistic κ. The value of k is defined throughout this paper by the relation

$$k = (n - 1)/(n - R) \tag{1}$$

where n is the number of cores in a given group and R is the vector resultant of the normalized magnetization vectors (Fisher 1953) in this group. The average value of θ at each site and the virtual pole position calculated from site means are also listed in Table II.

ANALYSIS OF RESULTS

It will be noted first that the mean direction of magnetization of the 53 stable cores is within 6 degrees of the mean direction given in Table I. This may be regarded as a relatively good confirmation of the previously available data, especially if we note that the precision estimate for the present group is relatively high. Another point worthy of mention is that, as a result of the cleaning treatment, the within-site precision estimates were noticeably increased in all cases except one.

A double χ^2 test was applied to check whether or not the magnetization directions of the 53 stable cores were distributed according to a Fisherian distribution. The first phase of this test consists in verifying whether or not the angle ψ between the individual direction of magnetization and the mean of the group has the density distribution suggested by Fisher (1953) as

$$df = \frac{\kappa e^{\kappa \cos\psi} \sin \psi \, d\psi}{2 \sinh (\kappa)} . \tag{2}$$

Letting the theoretical number of vectors within the ith interval ($\psi_i \leqslant \psi < \psi_{i+1}$)

TABLE II

Summary of palaeomagnetic data

Dyke	Site	N	Before cleaning				After cleaning at 150 Oe					Pole positions			
			n	k	D	I	n	k	D	I	Av. θ	Long.	Lat.	$\delta\chi$	$\delta\psi$
K	1	3	3	276	274	16	3	557	273	02	3.4	173 W	3	5	3
L	2	2	2	74	275	42	2	377	265	06	5.5	165	−1	13	6
I	3	4	4	1	267	58	4	57	272	00	5.8	173	2	12	6
I	4	5	5	68	269	34	5	315	268	15	3.3	165	4	4	2
J	5*	5	5	9	273	03	4	158	265	05	3.1	167	−2	7	4
J	6	3	3	233	271	28	3	279	269	12	4.1	166	3	7	4
C	7*	4	4	6	237	69	2	44	266	04	9.5	168	−1	38	19
E	8	5	5	89	262	19	5	195	262	−05	2.0	167	−7	5	3
D	9*	5	4	55	262	16	5	73	261	−01	3.1	165	−6	9	4
C	10	5	5	9	270	21	5	526	274	−08	3.4	177	−1	3	2
A	11†	2	2	1	083	−32	0	—	—	—	—	—	—	—	—
B	12	5	5	237	272	09	5	168	271	−07	3.0	174	−2	6	3
G	13*	4	4	16	117	66	0	—	—	—	—	—	—	—	—
H	14	3	3	264	267	07	3	2077	267	−10	3.6	173	−5	3	1
F	15	5	5	8	282	11	5	52	272	−13	5.0	177	−4	11	6
M	16	2	2	63	272	19	2	738	270	−09	9.2	174	−3	9	5
Mean		62	61	4	269	27	53	52	268	−01	4.1	171 W	−2	3	1

NOTE: N, number of cores collected; n, number of homogeneously magnetized cores; k, precision coefficient estimate (Fisher, 1953); D, clockwise angle with true north direction; I, positive downward inclination of magnetic vectors; * means that $(N\text{-}n)$ cores were found inhomogeneously magnetized; † means that one core appeared to have been magnetized by lightning; $\delta\chi$ and $\delta\psi$ are the semi axes the oval of confidence about of the poles at the .95 probability level.

TABLE III

Results of χ^2 test to verify Fisherian distribution

Upper limit of interval (degrees)	4	6	8	10	12	14	17	180			$\Sigma(O\text{-}E)^2/E$	$\chi^2_{.05}$	d
Expected number (E_i)	6.3	6.8	7.9	7.9	7.0	5.7	5.9	5.5					
Observed number (O_i)	5	3	10	11	7	8	5	4			5.6	9.5	4
Upper limit of interval (degrees)	3.6	72	108	144	180	216	252	288	324	360			
Expected number (E_j)	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3			
Observed number (O_j)	6	3	4	5	6	8	6	3	2	10	10.2	14.1	7

be E_i($E_i \geqslant 5$), and the observed number of vectors within the same interval be O_i, the summation

$$S = \sum_{i=1}^{p} \frac{(O_i - E_i)^2}{E_i} \tag{3}$$

is computed for the p intervals subdividing the range $0° \leqslant \psi \leqslant 180°$.

It is explained in textbooks on statistics (e.g. Bennett and Franklin 1954, p. 620) that if S be smaller than or equal to $\chi^2_{.05}$ with d degrees of freedom, there is no reason to accept at the 5% probability level the hypothesis that the observed set of data is not distributed according to the postulated theoretical distribution, which is that defined in equation (2) in the present case. The number of degrees of freedom is given by

$$d = p - 1 - g \tag{4}$$

where g stands for the number of constants to be estimated for the calculation of the values of $(E_i - O_i)$. It may be derived from (2) that

$$E_i = n(e^{\kappa(\cos\psi_i - 1)} - e^{\kappa(\cos\psi_{i+1} - 1)}). \tag{5}$$

Since κ and two of the direction cosines of the population mean must be estimated from the data in computing $(E_i - O_i)$, d is given by $(p - 4)$ in the present case. An estimate of κ is obtained from the data with the use of (1). The purpose of the second phase of the test is to verify if the observed directions of magnetization have a symmetrical distribution about their mean, as stipulated for the second condition of a Fisherian distribution. The mechanics of the test are the same as for the first phase. If r is the number of intervals subdividing the full range of 360° so that all E_j are greater than 5, the number of degrees of freedom is given this time by $(r - 3)$ because only the two direction cosines must be estimated from the data in computing the values of $(O_j - E_j)$. The results of the double test (Table III) show that there are no reasons to believe at the 5% probability level that the 53 directions of magnetization considered are not distributed according to Fisher's density function.

Denoting by k_a and k_b the precision estimates of samples a and b respectively, Watson (1956) has shown that if the ratio k_b/k_a $(k_b \geqslant k_a)$ is greater than the statistic $F_{2(N_a-1),\,2(N_b-1),\,\alpha}$, the directions of magnetization of the two samples most probably belong to populations characterized by different precision indexes. An application of this criterion to the values of k listed in Table II reveals that the 53 directions were not drawn from equally scattered populations.

A calculation of the overall within- and between-site precision coefficients w_0 and b_0 allows the application of another test suggested by Watson (1956). These coefficients are defined as:

$$w_0 = \frac{(N - S)}{N - \sum R_i}, \tag{6}$$

$$b_0 = \frac{(S-1)}{\sum R_i - R} \tag{7}$$

where N_i and R_i are respectively the number of cores and the vector sum of their normalized magnetization vectors for the ith site, S is the number of sites, and R is the resultant of the 53 normalized magnetization vectors. According to Watson's test, if the ratio w_0/b_0 is greater than the statistic $F_{2(S-1),\,2(N-S),\,\alpha}$ the site means are significantly distinct at the probability level α. It will be verified from the values obtained for w_0 and b_0 listed in Table IV that the 53 directions of magnetization are representative of more than one population of magnetization vectors at the .05 probability level.

It was shown in a previous paper (Larochelle 1966) that the angular measurement probable error is less than $\frac{1}{2}$ degree with the instrument and technique used in this study. As this is probably less than the average error introduced during the orientation of the cores *in situ*, the importance of this factor may be considered negligible for practical purposes. On the other hand, the orientation error may be assessed from the within-site precision estimates listed in Table II, which indicate that this error would likely not be much beyond 2 degrees. Most of the scattering in the magnetization directions of the 53 stable cores must therefore be attributed to a source independent of experimental procedures. On the other hand, it might be associated either with undetected differential tectonic movements within individual dykes, or with the fact that the dykes were intruded at different periods.

To investigate these two possibilities, the analysis was focussed on the three groups of data obtained for the cores collected from dykes I, J, and C respectively (Fig. 1). The within- and between-dyke precision estimates w_1 and b_1 as well as the within- and between-site precision estimates w_2 and b_2 were computed for these data (Table IV). The value of b_1 is significantly less

TABLE IV

Within- and between-site (dyke) precision indices

i	N	S	w_i	b_i	w_i/b_i	$F_{.05}$
0	53	14	127	19	6.8	1.62
1	23	3	78	14	5.6	2.61
2	23	6	140	18	7.8	2.08

NOTE: N is the number of cores incorporated in the calculations; S is the number of sites (dykes for $i = 1$) considered; $F_{.05}$ is the statistic $F_{2(S-1),2(N-S)}$ at the .05 probability level, extracted from tables.

than that of w_1 although the values of w_2 and of b_2 are respectively close to the values of w_0 and b_0 obtained for the 53 cores. These results suggest respectively (*a*) that the dykes were not magnetized simultaneously, (*b*) that the data obtained for the three dykes are representative of the 53 stable directions, and (*c*) that they have not been disturbed significantly since their emplacement.

Virtual pole positions representative of the 11 stably magnetized dykes sampled are plotted in Fig. 3. Unit weight was given to each of these points in

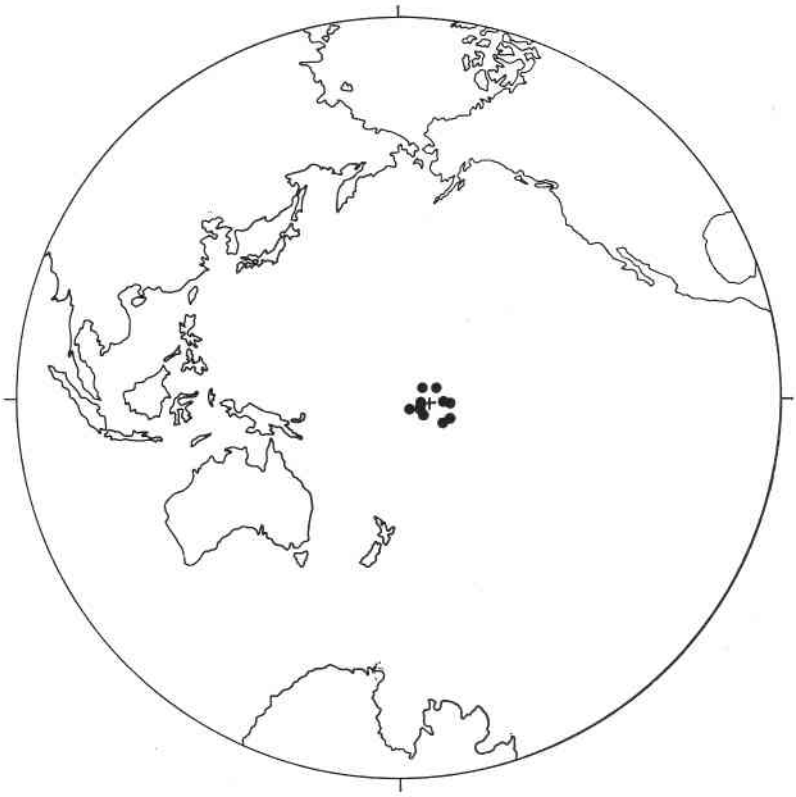


FIG. 3. Stereographic projection of pole positions derived from mean dyke directions of magnetization. The center of gravity of the group is represented by (+).

calculating their precision index estimate ($k = 214$), their angular standard deviation, $\Delta = 81°/\sqrt{k} = 5.5$ degrees, and their center of gravity (171° W., 2° S.). Whether or not this point should be considered as the geographic pole at the time the dykes were injected may now be investigated. It is known that if a diabase dyke sample be allowed to cool from say 600 °C in the present earth's field, the pole position that might be derived from its magnetization using the dipole equation would be closer to the present geomagnetic pole than to the geographic pole. If hypothetically, however, the operation should be repeated every year for the next 10 000 years, Cox and Doell (1960) suggest on the basis of the present-day configuration of the earth's field that the center of gravity of all these poles would probably correspond very closely with the present geographic pole. On the other hand, Creer (1962) and Cox (1962) have shown that the angular standard deviation Δ of the directions of magnetization would be a function of the magnetic latitude of the point where the experiment was performed. Finally Gough *et al.* (1964) have given evidence to support the hypothesis that the dispersion of the geomagnetic field with respect to an axial geocentric dipole field was similar to its present dispersion both in the early Precambrian and in Mesozoic times. The upper limit of Δ may be

calculated from the expression $\Delta_+ = \Delta\sqrt{F}$, where F is the statistic at the .05 probability level for $2(n-1)$, $2(n-1)$ degrees of freedom, n being here the number of dyke means considered. Thus, given an independent set of magnetization directions obtained from rocks collected at this magnetic latitude, it may be speculated that if their value of Δ_+ were significantly less than the Δ predicted for this latitude, the probability would be that the rocks acquired their magnetization within a period of time considerably shorter than 10 000 years. In applying this reasoning to the set of dyke mean directions, we must note first that the magnetic paleolatitude of the dykes is close to zero and that, accordingly, their angular standard deviation would be of the order of 19° (Creer 1962) if the injection of the dykes had spread over a period of 10 000 years or more. Since the actual value (8 degrees) obtained for Δ_+ is significantly less than the expected Δ, it is concluded that the dykes sampled were probably all injected with a few hundred or, at most, a few thousands of years. It is concluded further that the center of gravity of the poles plotted in Fig. 3 probably does not coincide with the geographic pole at the time of the dykes cooling, and that it would rather correspond to a virtual geomagnetic pole for a limited period of time during the Precambrian.

SUMMARY AND CONCLUSIONS

It is suggested from the present analysis and from previous data that the Sudbury area was at the magnetic equator at the time the dykes were formed, over 1 000 m.y. ago. Statistically controlled data indicate also that the dykes sampled for the present study were not disturbed appreciably since their original emplacement and that their cooling was a short duration event. It is suggested that all sampled dykes were intruded within considerably less than 10 000 years.

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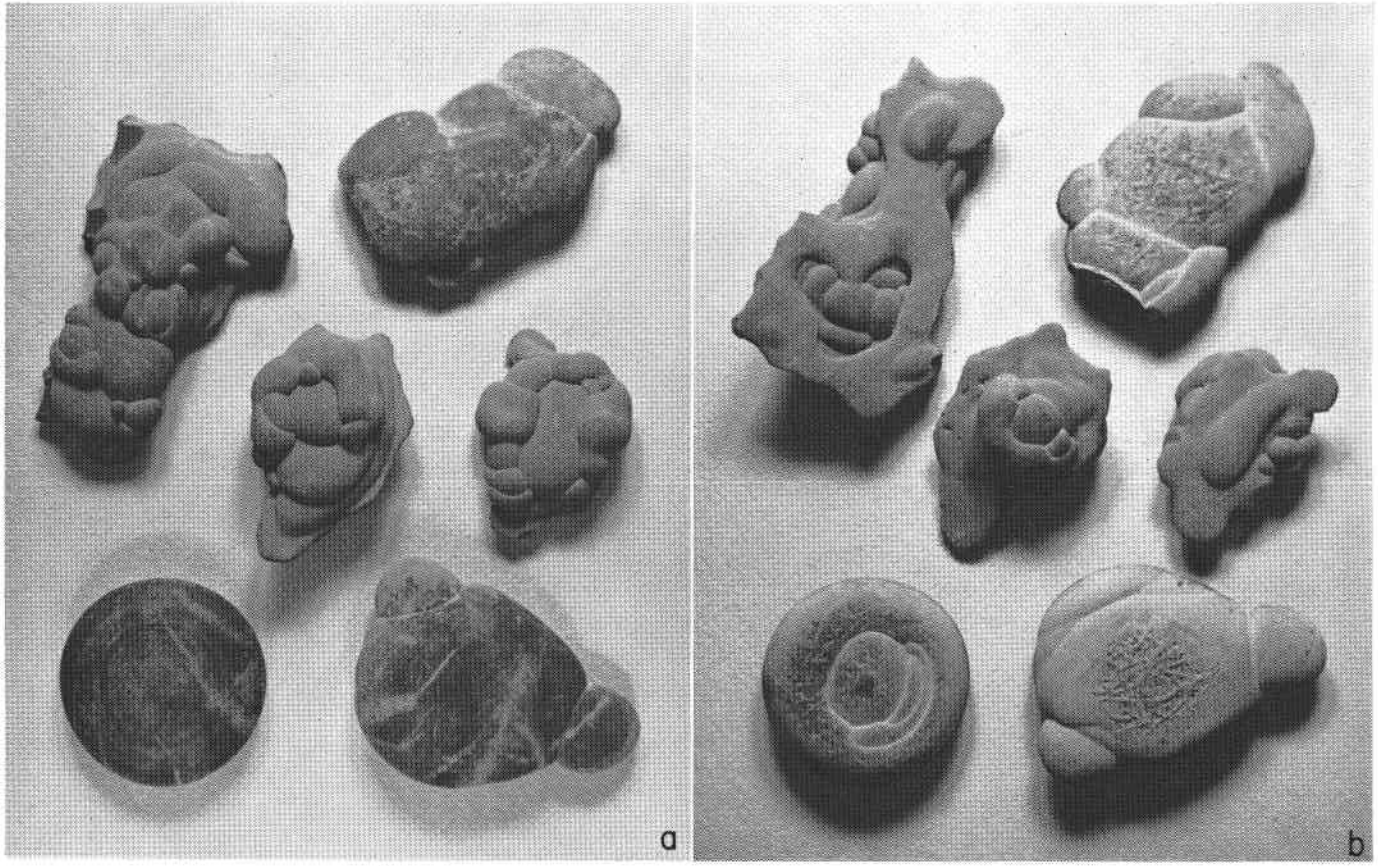


FIG. 1. Top (*a*) and bottom (*b*) views of concretions. The three concretions with uniform surfaces were from the Abitibi area, and the three concretions in the center of the photograph were from northwestern Vermont. The round concretion is 5 cm in diameter.